

Cracking the Mystery Function

PhET Function Builder · PhET

Years 7-10	~30 minutes	Mathematics · Algebra & Functions
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/function-builder>

Curriculum links: AC9M7A05 · AC9M8A04 · AC9M9A06 · AC9M10A05

Learning intentions

- I can test inputs and outputs to work out the rule of an unknown function.
- I can record input-output pairs in a table and describe the pattern algebraically.
- I can compare simple and multi-step function rules and explain how each step changes the output.

Getting started

Open PhET Function Builder and go to the Mystery screen. Choose one of the easier mystery function cards to start, feed in a few numbers, and note down what comes out before you try to guess the rule.

Tasks

1. Choose your first Mystery card. Feed in the input 1, then 2, then 3. Record the outputs you get.

2. Test at least four more inputs on the same mystery function and record the input-output pairs.

Input	Output

3. Based on your table, write down what you think the function rule is (e.g. 'add 3' or 'multiply by 2, then add 1').

4. Click to reveal the actual function machine. Were you correct? If not, what did you miss?

5. Choose a harder Mystery card with two steps. Test at least four inputs and record the input-output pairs.

Input	Output

6. Explain the strategy you used to work out the two-step function rule. What did you test first, and why?

7. Two different function rules can sometimes give the same output for one particular input, but not for others. Explain why testing just one input is not enough to be sure you have found the correct rule.

Extension (optional)

Choose the hardest Mystery card available. See if you can identify the rule using only three carefully chosen test inputs rather than four or five, and explain why the inputs you picked (e.g. 0, 1 and a larger number) helped you work it out efficiently.